Experiment #1: Introduction, Significant Figures & Errors

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Problems:

A: Significant figures and errors:

Q1: Determined the number of significant figures of the following:

1) 9.00 (3 S.f)
2) 0.000524 (3 S.f)
3) 1.00508 (6 S.f)
4) 0.008500 (4 S.f)
5) 8.0008090 (8 S.f).

Q2: Use (SF) Rules to find:

1) 5.49 + 3.9 + 0.269 - 2. = (8)
2) 6.256 × 3.08 = (19,3)
3) 61200. /9 = ($7 * 10^3$)

الحل الصحيح لسؤال الثالث في الصفحة القادمة

Q3: IF A=3.65 and B=1.98, then find $\Delta A = \Delta B = 0,01$

1) C ±ΔC, as (C=A-B). (1,67 ± 0,02)
2) C ±ΔC, as (C=A.B) (7,23 ± 0,06).
3) C ±ΔC, as (C=2A+AB) (14,5 ± 0,1).
4) C ±ΔC, as ($A^2.B^3$) (103 ± 2).

Q4: Student used a ruler that has a maximum rang (15 cm) to determine the length of table, which its length (2m = 200cm).so the uncertainty (error) in length is (ΔL = 0,7 cm).

$\frac{200\ cm}{15\ cm} = 13.\bar{3}$ → عدد مرات استخدام المسطرة لإيجاد طول الطاولة يساوي 13,3

$\Rightarrow \Delta L =$ دقة المسطرة * عدد المرات

$= 0,05\ cm * 13.\bar{3}$

$\Delta L = 0,7\ cm$

B: Graphs:

A student has made the following measurements for the sides of 6 cubes of some material with different volumes and its masses and tabulated the results in the following table.

(i) Use the table to find the volume and calculate the average mass density ($\rho = m/V$) of the material and its uncertainty.

$= \rightarrow \left| \frac{g_{ac} - g_{ex}}{g_{ac}} \right|$

Subject Date No.

(Q3) $A = 3.65$ & $B = 1.98$.

1) $C = A - B$

$= 3.65 - 1.98 = 1.67$

$\Delta C = \Delta A + \Delta B$

$= 0.01$

$C \pm \Delta C = 1.67 \pm 0.01$

2) $C = A \cdot B$

$= (3.65) \cdot (1.98) = 7.227$

$= 7.23$

$\Delta C = \left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \cdot C$

$= \left[\frac{0.005}{3.65} + \frac{0.005}{1.98} \right] \cdot 7.23$

$\Delta C = 0.028 \simeq 0.03$

$C \pm \Delta C = 7.23 \pm 0.03$

3) $C = 2A + AB$

$= 7.3 + 7.23 = 14.53$

$= 14.5$

$\Delta C = \frac{2(0.005)}{3.65} + (3.895 \times 10^{-3}) \cdot 14.5$

$\Delta C = 0.059$

$= 0.1$

$C \pm \Delta C = 14.5 \pm 0.1$

4) $C = A^2 \cdot B^3$

$= 103.4$

$= 103$

$\Delta C = \left[\frac{0.01}{3.65} + \frac{0.015}{1.98} \right] \cdot 103$

$\Delta C = 1.06$

$= 1$

$C \pm \Delta C = 103 \pm 1$

NOTEBOOK

(ii) Use the relation ($\rho=m/V$) to draw a suitable graph (V vs. m) on clean graph paper to find the mass density ρ from the slope, then calculate $\Delta\rho$. (write the answer as $\rho\pm\Delta\rho$)

	side length l(cm) Δl=0.05cm	*mass* m(g) Δm=0.05g	*volume* V(cm^3)	ΔV
①	3.3	258.4	36	$26\ \frac{\Delta V}{V}=0,05$
②	4.8	728.0	$11*10^1$	$31\ \frac{\Delta V}{V}=0,03$
③	6.2	1675.6	$24*10^1$	$61\ \frac{\Delta V}{V}=0,02$
④	8.7	4515.8	$66*10^1$	$116\ \frac{\Delta V}{V}=0,02$
⑤	9.5	5743.2	$86*10^1$	$146\ \frac{\Delta V}{V}=0,02$

$\rho\pm\Delta\rho = 7,2\pm 0,4$

$\rho\pm\Delta\rho = 6,62\pm 0,12$

$\rho\pm\Delta\rho = 6,98\pm 0,14$

$\rho\pm\Delta\rho = 6,84\pm 0,14$

$\rho\pm\Delta\rho = 6,68\pm 0,13$

$$\Delta\rho = \rho\left(\frac{\Delta m}{m}+\frac{\Delta V}{V}\right)$$

① $\Delta\rho = 7,2\left(\frac{0,05}{258.4}+0,05\right)$

$\Delta\rho_{(1)} = 0,4$

② $\Delta\rho = 6,62\left(\frac{0,05}{728}+0,03\right)$

$\Delta\rho_{(2)} = 0,12$

③ $\Delta\rho = 6,98\left(\frac{0,05}{1675.6}+0,02\right)$

$\Delta\rho_{(3)} = 0,14$

④ $\Delta\rho = 6,84\left(\frac{0,05}{4515,8}+0,02\right)$

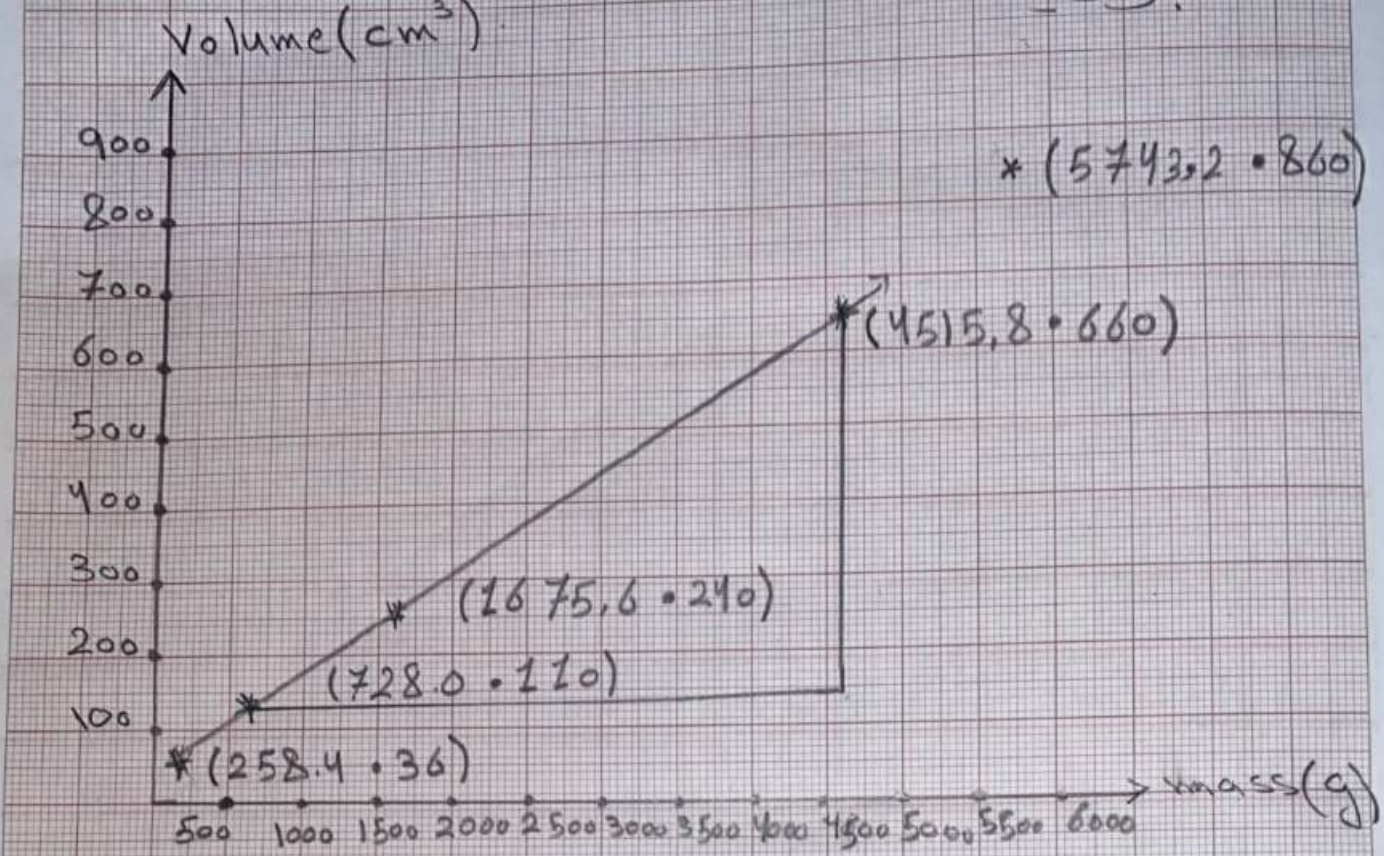
$\Delta\rho_{(4)} = 0,14$

⑤ $\Delta\rho = 6,68\left(\frac{0,05}{5743.2}+0,02\right)$

$\Delta\rho_{(5)} = 0,13$

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$$* \text{slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta \text{Volume}}{\Delta \text{mass}}$$

$$\text{slope} = \frac{V_2 - V_1}{m_2 - m_1}$$

$$\text{slope} = \frac{660 - 110}{4515.8 - 728.0} = 0.145 \ cm^3/g$$

$$\rho = \frac{1}{\text{slope}} = \frac{1}{0.145} = 7 \ g/cm^3$$

$$\rho = 7 \ g/cm^3$$

$$\Delta\rho = \rho\left(\frac{\Delta m}{m} + \frac{\Delta V}{V}\right)$$

$$= 7\left(\frac{0.05}{728} + 0.03\right)$$

$$\Delta\rho = 0.2 \ g/cm^3$$

$$\rho \pm \Delta\rho = 7 \pm 0.2$$

Experiment #2: Measurements

1. Do all the calculations required to fill in the data tables. Make sure you round off your answers to the proper number of significant figures and give the units of all measured and calculated quantities.

Data Table (1): Rectangular Wooden Block			
Length, L	120,8 mm	ΔL	0,05
Width, W	40,7 mm	ΔW	0,05
Height, H	29,2 mm	ΔH	0,05
Surface Area, A	19280 mm^2	ΔA	$4,7 * 10^1$
Volume, V	$144 * 10^3$ mm^3	ΔV	$5 * 10^2$
Mass, M	91,90 g	ΔM	0,01
Density, ρ	$0,6382 * 10^{-3}$ g/mm^3	$\Delta\rho$	$0,002 * 10^{-3}$ g/mm^3

$\frac{\Delta A}{A} = 0,00244$

$\frac{\Delta V}{V} = 0,0038$

$= 0,6382\ g/cm^3$

$= 0,002\ g/cm^3$

Data Table (2): Cylinder			
Diameter, d	12,60 mm	Δd	0,005
Radius, $r = d/2$	6,30 mm	Δr	0,002
Height, H	189 mm	ΔH	0,5
Surface Area, A	7729 mm^2	ΔA	20 mm^2
Volume, V	$236 * 10^2$ mm^3	ΔV	$8 * 10^1$
Mass, M	184,54 g	ΔM	0,01
Density, ρ	$0,78195 * 10^{-2}$ g/mm^3	Δρ	$0,003 * 10^{-2}$ g/mm^3

Data Table (3): Metallic Ball			
Diameter, d	15,78 mm	Δd	0,005
Radius, $r = d/2$	7,89 mm	Δr	0,002
Surface Area, A	782 mm^2	ΔA	0,2
Volume, V	$206 * 10^1$ mm^3	ΔV	2
Mass, M	16,31 g	ΔM	0,01
Density, ρ	$0,792 * 10^{-2}$ g/mm^3	Δρ	$0,001 * 10^{-2}$ g/mm^3

Calculations:

Show sample calculation, for example, the complete calculation for Surface Area.

* تم النقل على ورقة خارجية

$A = 2LW + 2WH + 2LH$

$= 2(492) + 2(119*10^1) + 2(353*10^1)$
$*10^1$

$A = 19280\ mm^2$.

$L \pm \Delta L = 120,8 \pm 0,05$

$W \pm \Delta W = 40,7 \pm 0,05$

$H \pm \Delta H = 29,2 \pm 0,05$



$= 2Lw\left(\left(\frac{\Delta L}{L}\right) + \frac{\Delta w}{w}\right)\right) + 2wH\left(\frac{\Delta W}{W} + \frac{\Delta H}{H}\right) + 2LH\left(\frac{\Delta L}{L} + \frac{\Delta H}{H}\right)$

$= 2(492*10^1)\left(\frac{0,05}{120,8} + \frac{0,05}{40,7}\right) + 2(119*10^1)\left(\frac{0,05}{40,7} + \frac{0,05}{29,2}\right) + 2(353*10^1)\left(\frac{0,05}{120,8} + \frac{0,05}{29,2}\right) =$

$2*10^1 \quad + 0,7*10^1 + \quad 2*10^1 =$

$\Delta A = 4,7*10^1\ mm^2$

$V = L * W * H$

$= 120,8 * 40,7 * 29,2$

$= 144*10^3\ mm^3$

$\Delta V = V\left(\frac{\Delta L}{L} + \frac{\Delta W}{W} + \frac{\Delta H}{H}\right)$

$= 144*10^3\left(\frac{0,05}{120,8} + \frac{0,05}{40,7} + \frac{0,05}{29,2}\right)$

$\Delta V = 5*10^2$

$\rho = \frac{M}{V}$

$= \frac{91,90}{144*10^3} = 6,382*10^{-4}$

$\Delta \rho = \rho\left(\frac{\Delta m}{m} + \frac{\Delta V}{V}\right)$

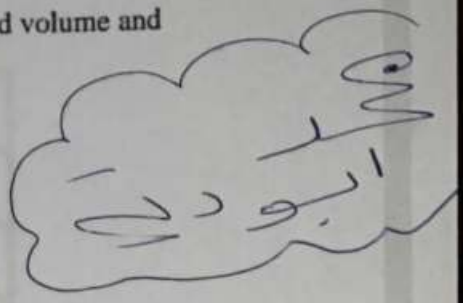
$= 0,6382*10^{-3}\left(\frac{0,01}{91,90} + 0,003\right)$

$\Delta \rho = 2*10^{-6}\ g/mm^3$

Results:

Provide the error formula used as well as the calculation of the errors of Area and volume and density in in the previous table.

Table (1)



$$A = 2Lw + 2wH + 2LH$$

$$\Delta L = 2Lw\left(\frac{\Delta L}{L} + \frac{\Delta w}{w}\right) + 2wH\left(\frac{\Delta w}{w} + \frac{\Delta H}{H}\right) + 2LH\left(\frac{\Delta L}{L} + \frac{\Delta H}{H}\right)$$

(2) $V = LwH$

$$\Delta V = V\left(\frac{\Delta L}{L} + \frac{\Delta w}{w} + \frac{\Delta H}{H}\right)$$

(3) $\rho = \frac{m}{V}$

$$\Delta \rho = \rho\left(\frac{\Delta m}{m} + \frac{\Delta V}{V}\right)$$

Conclusion:

1. We learned the difference between measurements and error ratio for both ruler,
   - Vernie and micro.



2. The accuracy of measurement varies from person to person.

Experiment #3: Vectors

Data:

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Table 1: Force table

Mass	Vector	Magnitude	Direction
M_1= 0,155	F_1	1,55	12°
M_2= 0,195	F_2	1,95	135°
M_3= 0,170	F_3	1,70	266°

Table 2: Components method

Vector	X-component	Y-component
F_1	1,52	0,322
F_2	-1,38	1,38
F_3	-0.12	-1,696
	ΣF_x 0,02	ΣF_y

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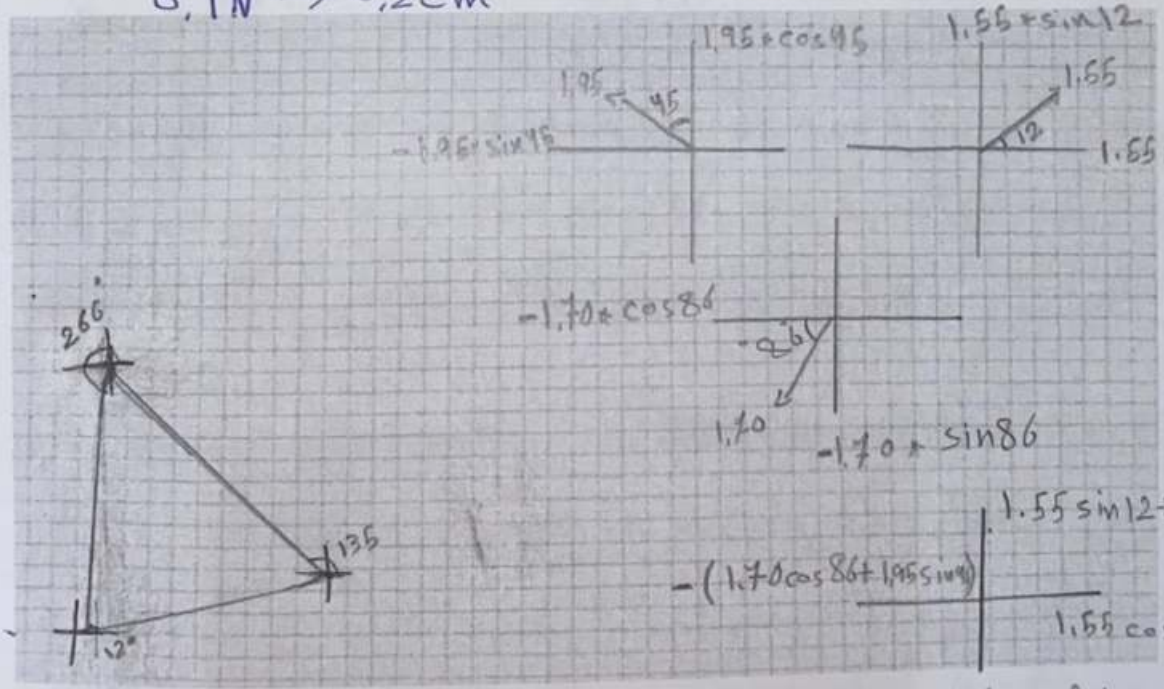
0,008 ≈ 0,01

R= 0,0223 ≈ 0,02 , θ_R= 26,57°

Graphical method:

Scale: 0,1N → 0,2cm

R= 0,02, θ_R= 26,57°



ΔE = 0,02cm

R_x = 0,02

R_y = 0,01

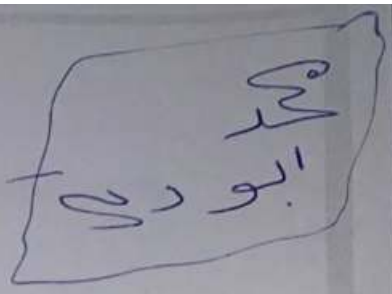
$R = \sqrt{(0,02)^2 + (0,01)^2}$ =

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$ =

$F = mg$ → $F_1 = 1.55$

$F_2 = 1.95$

$F_3 = 1.70$



Calculations:

Show calculation for R_1, R_2, and R_3.

$F_{1x} = F_1 \times \cos 12$ $\quad F_{2x} = F_2 \times \cos 135$ $\quad F_{3x} = F_3 \times \cos 266$

$F_{1x} = 1.52$ $\quad F_{2x} = -1.38$ $\quad F_{3x} = -0.12$

$F_{1y} = F_1 \times \sin 12$ $\quad F_{2y} = F_2 \times \sin 135$ $\quad F_{3y} = F_3 \sin 266$

$F_{1y} = 0.322$ ✓ $\quad F_{2y} = 1.38$ ✓ $\quad F_{3y} = -1.696$ ✓

$R_1 = 1.55$ $\quad R_2 = 1.95$ $\quad R_3 = 1.7$

$\vec{|F|} = R = \sqrt{(0.02)^2 + (0.01)^2}$

$R = 0.02$

$\Theta = \tan^{-1}\left(\frac{0.01}{0.02}\right)$

$\Theta = 26.57°$

$F_x = F_1 \cos 12 + F_2 \cos 135 + F_3 \cos 266$

$F_x = 0.02$

$F_y = F_1 \sin 12 + F_2 \sin 135 + F_3 \sin 266$

$F_y = 0.01$

$F_3 = \sqrt{(F_1)^2 + (F_2)^2 + 2F_1F_2\cos(\theta_2 - \theta_1)} = \sqrt{(1.55)^2 + (1.95)^2 + 2 \times 1.55 \times 1.95 \times \cos 123}$

My work is right ← $F_3 = 1.706 \approx 1.7$ ✓

Discussion:

Through my use of the methods used to achieve results, it turns out that the results are similar and that the force analysis has a rather higher error rate.

Conclusion:

1. Using the vector addition method to find the result is more accurate than using the force analysis method.

2. I learned more than one way to find the result.

Experiment #7: Uniform Circular Motion

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Data:

Radius (**r**) = 0,3 m, rubber bob mass (**m**) = 0,04 kg

Table (1)

0.012

Hanging mass M(Kg)	F_c=Mg (N)	Time for 15 rev. (t_{15}) (s)	Period $T=t_{15}/15$	Speed, v (m/s)	v^2 (m^2/s^2)	Frequency, f=1/T	f^2 ($1/s^2$)
0,03	0,3	7.64	0,51	3.7	13.7	1,96	3,84
0,04	0,4	7,46	0,5	3,8	14.4	2	4
0,05	0,5	6,84	0,46	4.1	16,8	2.17	4,71
0,06	0,6	6,08	0,41	4,6	21,2	2,44	5,96
0,07	0,7	5,88	0,4	4,7	22,1	2.5	6,25

Discussion and analysis:

1) Complete Table 1. ✓
2) Plot F_c versus v^2. ✓ → (mg) Vs (f^2)
3) Find the physical meaning and the magnitude of the slope on the graph paper. ✓
4) Using the slop, find the magnitude of (**m**). ✓
5) The percentage error of (**m**). ✓
6) Write down the answers to parts (3-5) in the Results Section ✓

$(mg)=4\pi^2 r m(f^2)$

Results:

3) Physical meaning:

$slop=4\pi^2 r^2 m$

$slop=\frac{y_2-y_1}{x_2-x_1}$

slope = 0,166

4)

$slope=4\pi^2 r m$

$m=slope/4\pi^2 r$

$=\frac{0,166}{4\times\pi^2\times 0,3}$

m = 0,014

5)

$P.E=\left|\frac{m_g-m_{act}}{m_{act}}\right|\times 100\%$

$=\left|\frac{0,04-0,012}{0,012}\right|\times 100\%$

P.E = 16,7 %

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B. Finding rubber bob mass by keeping the hanging mass constant, and changing the spinning radius.

Repeat the Procedure from (1-9) in part A, but by Keeping the hanging mass constant, and changing the spinning radius, and then Record your data in Table 2.

Data:

Hanging mass (M) = 0,05 kg, rubber bob mass (m) = 0,04 kg

0.012

Table (2)

Radius, r (m)	Time for 15 rev. (t_{15}) (s)	Period $\tau=t_{15}/15$	τ^2 (s^2)	v^2 (m^2/s^2)	Frequency, $f=1/T$
0,3	6,84	0,456	0,208	17,1	2.19
0,35	7,42	0,495	0,245	19,7	2,02
0,4	7,67	0,511	0,261	24,2	1,96
0,45	8	0,533	0,284	28,1	1,88
0,5	9,65	0,643	0,413	23,8	1,56

Discussion and analysis:

1) Complete Table 2.
2) Plot **r** versus T^2.
3) Find the physical meaning and the magnitude of the slope on the graph paper.
4) Using the slope, find the magnitude of (**m**).

عبد ابو دية

5) The percentage error of (m). ✓

6) Write down the answers to parts (3-5) in the Results Section. ✓

Results:

$$Mg = \frac{4\pi^2 r m}{T^2}$$

3) Physical meaning : slope $= \frac{M g}{4\pi^2 m}$

4) slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{0,5 - 0,3}{0,413 - 0,208}$

slope $= 0,976$

$$m = \frac{M g}{4\pi^2 \times \text{slope}} \Rightarrow m = \frac{0,05 \times 10}{4 \times \pi^2 \times 0,976}$$

$m = 0,013$ Kg

$$P.E = \left|\frac{m_{th} - m_{act}}{m_{act}}\right| \times 100\%$$

$$= \left|\frac{0.013 - 0,012}{0,012}\right| \times 100\%$$

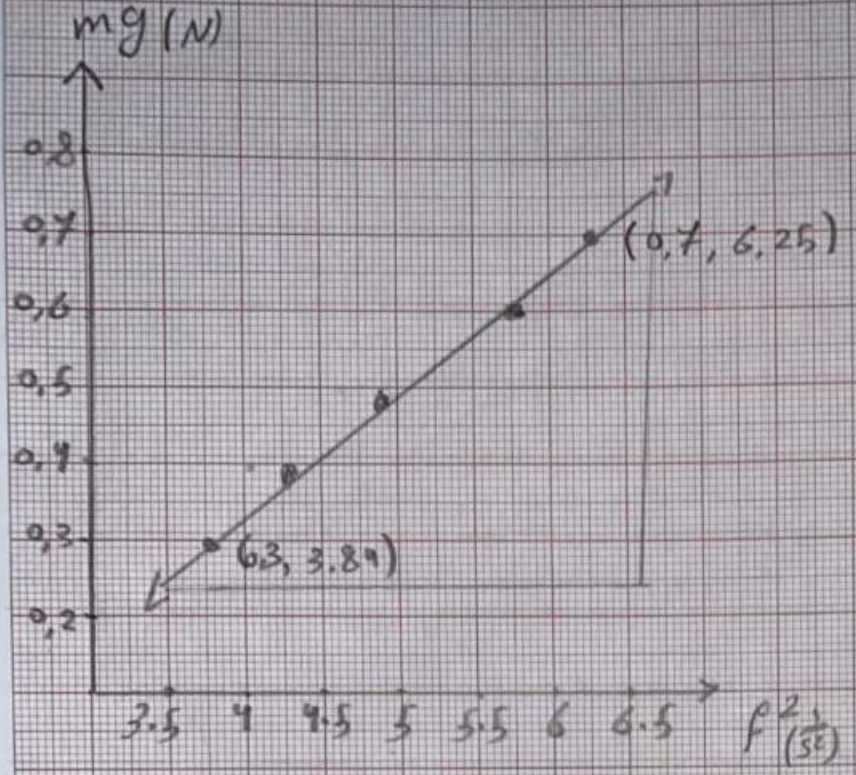
$P.E = 8,34\%$

Conclusion:

1. The velocity of uniform motion remains constant i.e. unchanged.

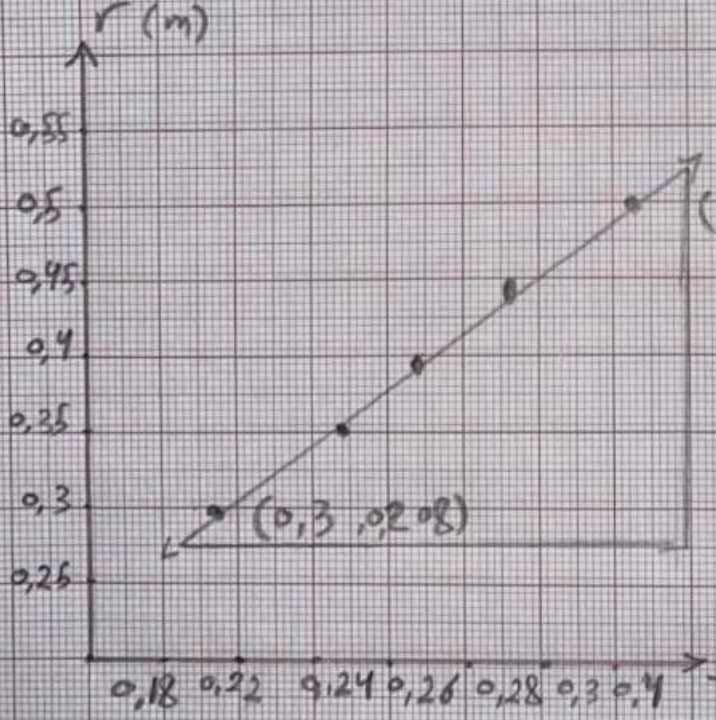
2- The net force on the object is also in the radially inward direction. This force is called central beetle force.

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(Table (1))

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0,7 - 0,3}{6.25 - 3.84} = 0,014$$



(Table (2))

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0,5 - 0.3}{0.413 - 0,208} = 0,013$$

Experiment #4: Newton's second law

Part 1: (When the total mass is constant):

Increase the values of (m_h) by the same amount decrease (M_c)

(Keep the total mass constant).

***Data*:**

M_c+m_h (total mass) = 0,285 kg, S= 0,5 m

Note that:-

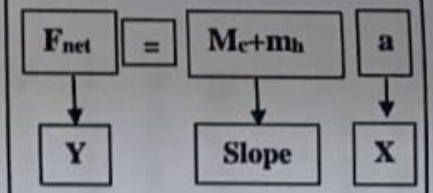


Table (1)

m_h(kg)	t (s)	F_{net}(N)	a (m/s^2)
0,01	2,00128	0,1	0,25
0,03	1,10235	0,3	0,82
0,05	0,77513	0,5	1,66
0,07	0,64328	0,7	2,42
0,09	0,51926	0,9	3,71

Note that:

$$F_{net} = (M_c + m_h)\, a$$

$$m_h \cdot g = (M_c + m_h) \cdot \frac{2.S}{t^2}$$

(m_h) and (t) are inversely proportional

- **Discussion and analysis:**

1) Fill Table (1) by moving one of the movable masses from (M_c) to (m_h),to keep the total mass is constant, and (F_{net}) will increase in each try.
2) Plot a graph between (F) and (a).

3) Calculate the slope.
4) Determine the physical meaning of the slope.
5) Find the (M_c+m_h) expected value.
6) Find the percentage error for the total mass.

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Part 2: (when the driving force (F_{net}) is constant)

In this part, we will let (m_h) constant, so the driving force will be constant.

To get two pairs for plots, we will increase (Mw)

In every attempt, so (a) will change in every attempt.

Note that:

$$a = F_{net} \cdot \frac{1}{M_w + m_h}$$

Y | Slope | X

Data:

$F_{net\ (true\ value)} = m_h.g =$ 0,2 N

Table (2)

M_w(kg)	(M_w+m_h) (kg)	1/ (M_w+m_h)(kg^{-1})	t (s)	a (m/s^2)
0,195	0,215	4.65	1,19546	0,7
0,215	0,235	4.26	1,24879	0,64
0,235	0,255	3.92	1,32658	0,57
0,255	0,275	3.64	1,39580	0,51
0,265	0,285	3.5	1,46707	0,46

- **Discussion and analysis:**
  1) Fill the table above, by increasing (Mw) in each trial.
  2) Plot the relation between (a) and ($1/ M_w+m_h$).
  3) Calculate the slope.
  4) Determine the physical meaning of the slope.
  5) Find the expected value of net force.
  6) Find the percentage error for the net force.

Note that:

$a = F_{net} \cdot (\frac{1}{M_w+m_h})$

$\frac{2S}{t^2} = F_{net} \cdot \frac{1}{M_w+m_h}$

(M_w) and (t) are directly proportional

Results:

Write down the answers to all questions above in both parts here:

Table (1).

* slope $= \frac{\Delta y}{\Delta x} = \frac{0,7 - 0.1}{2,42 - 0,25} \approx 0,280$

* physical meaning $= M_w + M_h = 0,28$

* $(M_w + M_h)_{(expected)} = 0,285 Kg.$

* $P.E = \left|\frac{0,285 - 0,280}{0,285}\right| * 100\% = 2\%$

P.E = 2%

Table (2).

* slope $= \frac{\Delta y}{\Delta x} = \frac{0,7 - 0,51}{4.65 - 3.64} = 0,19$

* Physical meaning $= F_{net} = 0,19$

* $F_{net\ (expected)} = 0,2$

* $P.E = \left|\frac{0,2 - 0,19}{0,2}\right| * 100\% = 5\%$

P.E = 5%

- **Conclusion**

1. The acceleration of an object is directly related to the net force and inversely to its mass.

2. It shows that the motion is accelerated only when force is applied.

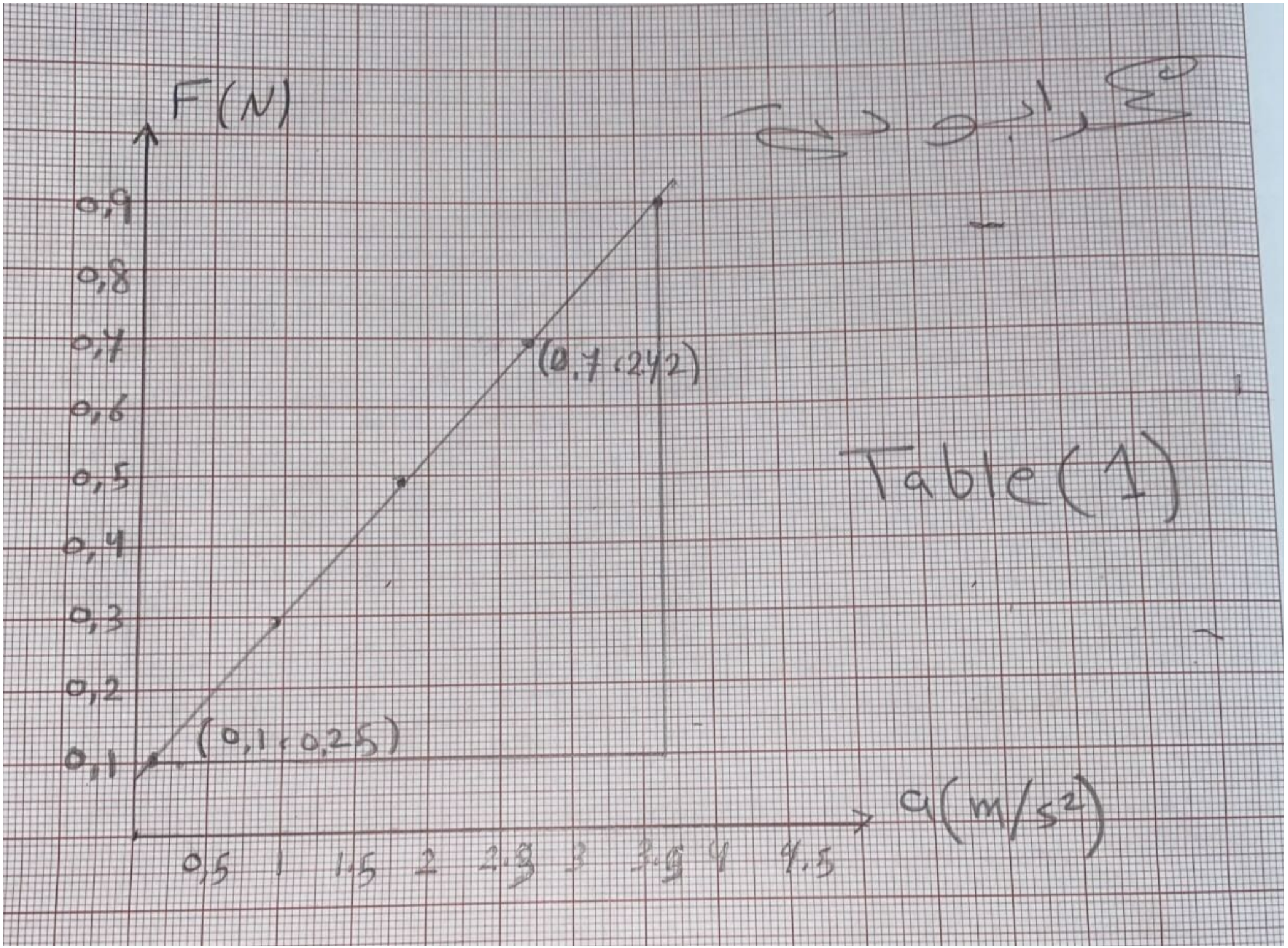


Table (1)

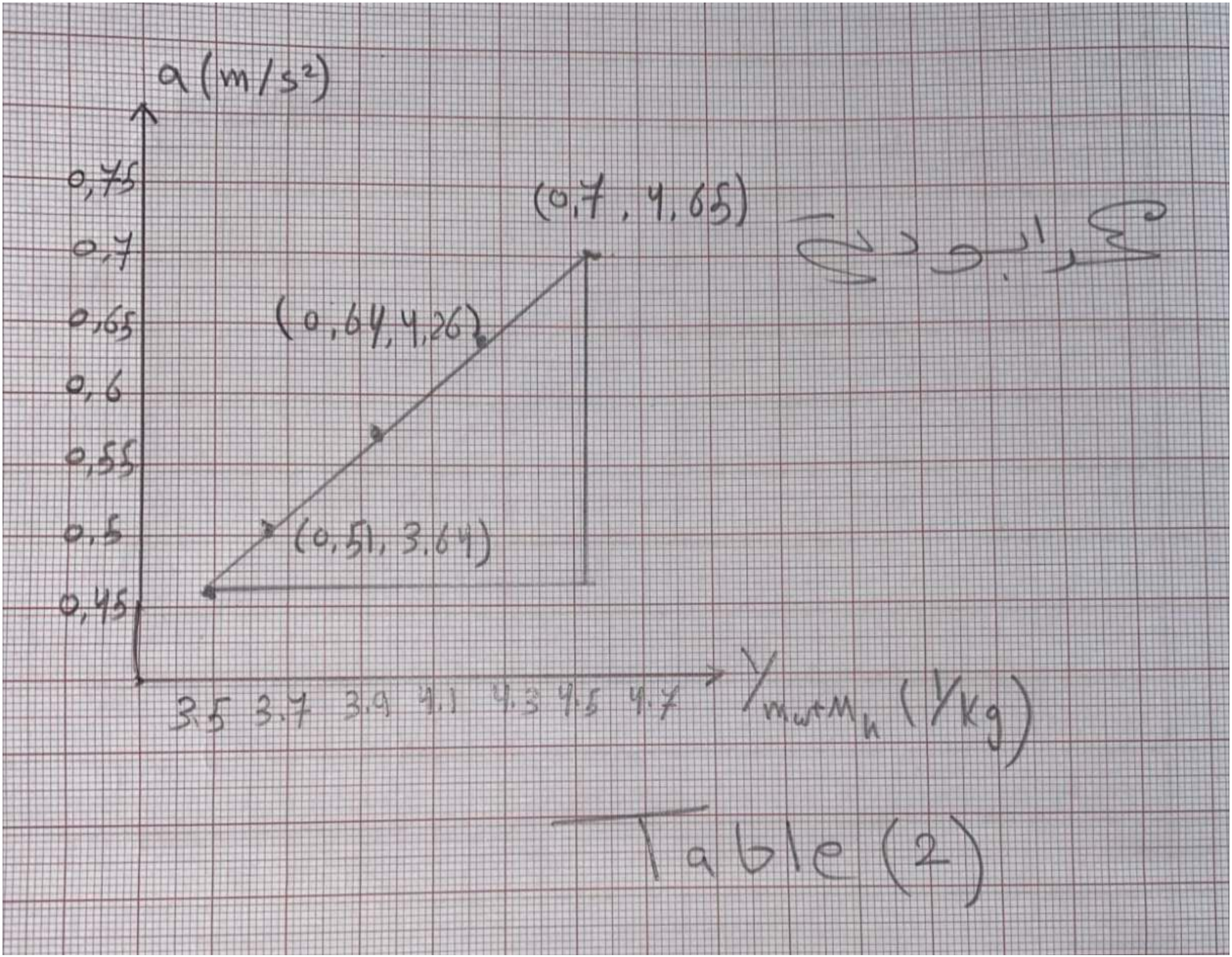

a (m/s²)
0,75
0,7
0,65
0,6
0,55
0,5
0,45
(0,7 , 4,65)
(0,64, 4,26)
(0,51, 3,64)
3.5 3.7 3.9 4.1 4.3 4.5 4.7
1/m+M (1/kg)

Table (2)

Experiment #5: Gravitational potential energy

Part 1: Determining the acceleration due to gravity experimentally. (Fixing (h_i) and changing (m)).

- **Data:**

h_i=0.3 (m), M=0,195 (kg), L=0,025 (m)

m(kg)	t(s)	V_f=L/t (m/s)	v_f^2(m^2/s^2)	$K_{f(M,m)}$ =1/2.(M+m) v_f^2 (J)
0,01	0,0453	0,55	0,302	0,030955
0,02	0,033	0,76	0,578	0,062135
0,03	0,027	0,92	0,846	0,095175
0,04	0,024	1,04	1,08	0,1269
0,05	0,022	1,14	1,3	0,15925

- **Discussion and analysis:**

1) Measure (h_i) which you want to fix, record the mass of (M), and record the length of the flag (L). ✓
2) Fill the table below by changing (m) and fixing (h_i). ✓
3) On the graph paper plot ($K_{f(M,m)}$) versus (m) ✓
4) Write the physical meaning of the slope. ✓
5) Find the slope. ✓
6) Find the acceleration of gravity (g). ✓
7) Find the percentage error in (g). ✓
8) Write down the answers to parts (4-7) in the Results Section below ✓

Note that:

$K_{f(m,M)} = g.h_i \cdot m$

$Y = SLOPE \cdot X$

Results:

(4) The physical meaning = g.h

$slope = \frac{\Delta y}{\Delta x} = \frac{0,1269 - 0,030955}{0,04 - 0,01}$

(5) slope = 3.18

(6) $g = \frac{slope}{h} = 10,6\ m/s^2$

$P.E = \left|\frac{10 - 10.6}{10}\right| * 100\%$

(7) P.E = 6 %

Part 2: Find the acceleration of gravity experimentally. (Fixing (m) and changing (h_i)).

- **Data**

L = 0,025 (m) m = ~~0,1~~ 0,02 (kg), M = 0,195 (kg)

h_i (m)	t (s)	$V_f = L/t$ (m/s)	$K_{f(M,m)} = 1/2(M+m)\, v_f^2$ (J)
0,1	0,057	0,44	0,021
0,2	0,041	0,61	0,04
0,3	0,033	0,76	0,062
0,4	0,029	0,86	0,08
0,5	0,026	0,96	0,1

- **Discussion and analysis:**

1) Detect (m) which you want to fix, record the mass of (M), and record the length of the flag (L).
2) Fill the table by changing (hi) and fixing (m).
3) On the graph paper plot ($K_{f(M,m)}$) versus (h_i).
4) Write the physical meaning of the slope.
5) Find the slope.
6) Find the acceleration of gravity (g).
7) Find the percentage error in (g).
8) Write down the answers to parts (4-7) in the Results Section

Note that:

$K_{f(m,M)} = g.m \cdot h_i$

$Y = SLOPE \cdot X$

Results:

(4) The phy. meaning = g · m

(5) $slope = \frac{0,1 - 0,021}{0,5 - 0,1} = 0,1975$

$g = \frac{slope}{m} = \frac{0,1975}{0,02}$

(6) $g = 9.9\ m/s^2$

(7) $P.E = \left|\frac{10 - 9.9}{10}\right| \times 100\% = 1\%$

- **Conclusion:**

(1) The sum of the potential energy and kinetic energy at any position in a closed system will always remain the same.

(2) The total energy during the motion does not change, but is constant.

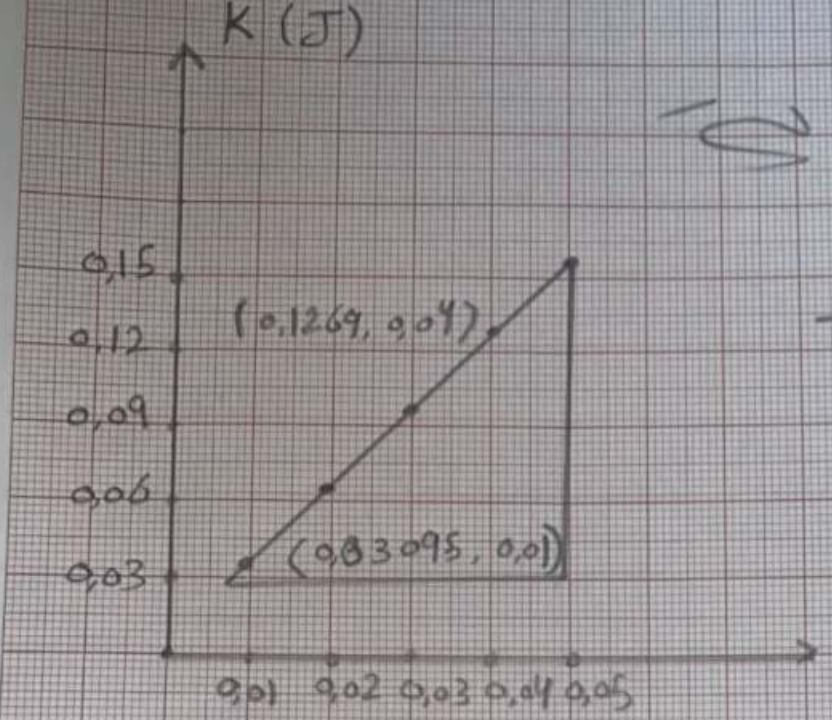


Table (1)

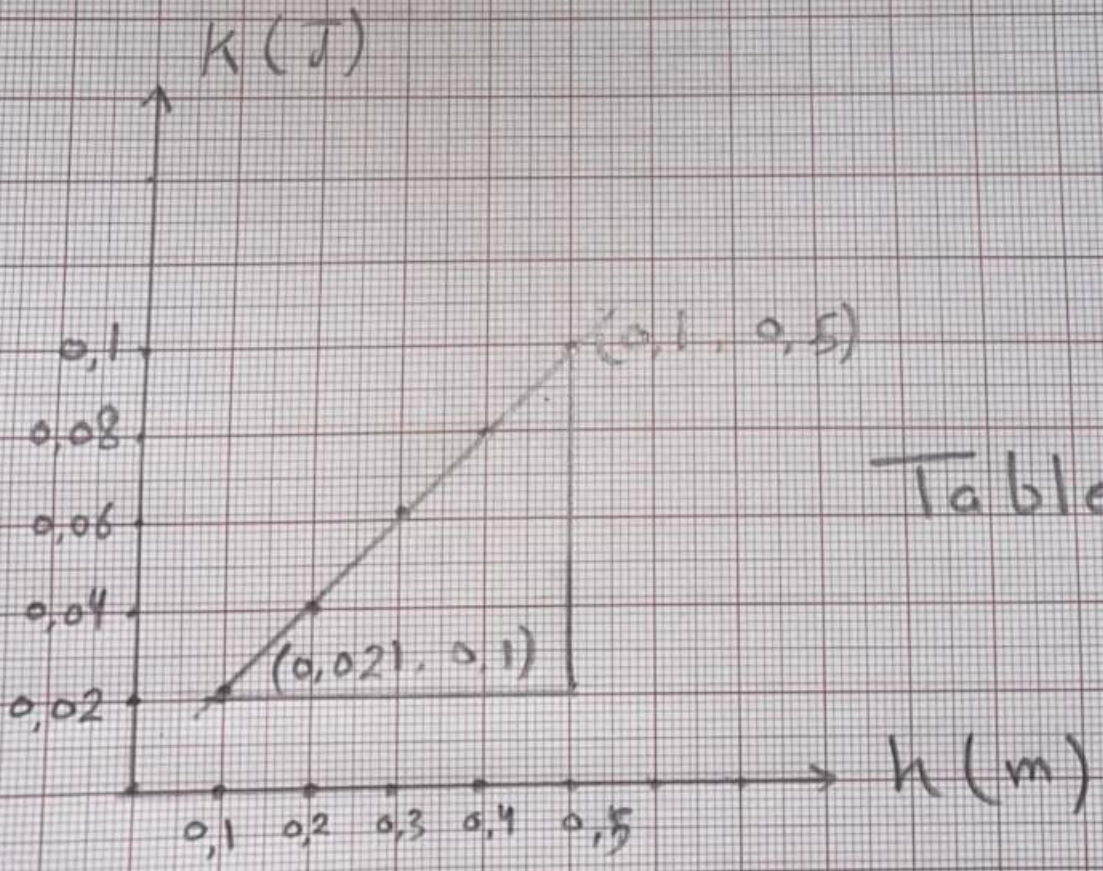


Table (2)

Experiment #6: ONE-DIMENSIONAL COLLISIONS

A. Elastic collision

L_1=0,025 (m) L_2=0,025 (m)

Before collisions								
Trial	M_1 (Kg) Cart 1	M_2 (Kg) Cart 2	T_{1i} (s)	T_{2i} (s)	$V_{1i}=L_1/T_1$ (m/s)	$V_{2i}=L_2/T_2$ (m/s)	Pi (Kg.m/s)	KE_i (J)
1	0,195	0,195	0,05846	0,07094	0,4276	-0,3524	0,014664	0,029935
2								

After collisions								
Trial	M_1 (Kg) Cart 1	M_2 (Kg) Cart 2	T_{1f} (s)	T_{2f} (s)	$V_{1f}=L_1/T_1$ (m/s)	$V_{2f}=L_2/T_2$ (m/s)	P_f (Kg.m/s)	KE_f (J)
1	0,195	0,195	0,05499	0,07379	0,45463	-0,338799	0,022587	0,031349
2								

Discussion and Analysis:

محمد ابو دية

1. Is the Linear momentum conserved? Explain that numerically.

Yes, it is.

$P_i = P_{i_1} + P_{i_2}$

$= m_1 V_1 + m_2 V_2$

$= (0,195 * 0,4276) + (0,195 * -0,3524)$

$P_i = 0,014664 \text{ Kg.m/s}$

$P_p = (0,195 * 0,45463) + (0,195 * -0,338799)$

$P_f = 0,022587 \text{ Kg.m/s}$

$\Delta P = P_f - P_i$

$= 0,022587 - 0,014664$

$\Delta P = 0,008193 \text{ kg.m/s} \simeq 0$ ✓

2. Is the kinetic energy conserved? Explain that numerically.

Yes, it is.

$KE_i = KE_{i_1} + KE_{i_2}$

$= \frac{1}{2} m_{i_1} V_{i_1}^2 + \frac{1}{2} m_{i_2} V_{i_2}^2$

$= (\frac{1}{2} * 0,195 * (0,4276)^2) + (\frac{1}{2} * 0,195 * (0,3524)^2)$

$KE_i = 0,029935 \text{ J}$

$\Delta KE = KE_f - KE_i$

$= 0,031344 - 0,029935$

$\Delta KE = 0,0014 \text{ J} \simeq 0$ ✓

محمد ابو دية

$KE_f = \frac{1}{2} m_{f_1} V_{f_1}^2 + \frac{1}{2} m_{f_2} V_{f_2}^2$

$= (\frac{1}{2} * 0,195 * (0,45463)^2) + (\frac{1}{2} * 0,195 * (0,338799)^2)$

$KE_f = 0,031344 \text{ J}$

عبد ابو دية

B. Perfectly inelastic collision

L_1= 0,025 (m) L_2= 0,025 (m)

Before collisions								
Trial	M_1 (Kg) Cart 1	M_2 (Kg) Cart 2	T_{1i} (s)	T_{2i} (s)	$V_{1i}=L_1/T_1$ (m/s)	$V_{2i}=L_2/T_2$ (m/s)	Pi (Kg.m/s)	KE_i (J)
1	~~0,195~~	~~0,195~~	~~0,0992~~	~~X~~	~~0,251864~~	~~X~~	~~0,098774~~	~~0,0251113~~
2	0,195	0,195	0,04926	X	0,507511	X	0,098965	

0,025113

After collisions											
Trial	M_1 (Kg) Cart 1	M_2 (Kg) Cart 2	M_f (Kg)	T_{1f} (s)	T_{2f} (s)	T_f (s)	$V_{1f}=L_1/T_1$ (m/s)	$V_{2f}=L_2/T_2$ (m/s)	$V_f=L/T_f$ (m/s)	P_f (Kg.m/s)	KE_f (J)
1	0,195	0,195	0,39	0,08258			-0,30296	0,303324	0,30314	0,11822	0,017719
2											

0,08247 = 0,16494 / 2

0,08242

Data Analysis:

1. Is the Linear momentum conserved? Explain that numerically.

yes, it is ..

$P_i = m_i V_i$

$= 0,195 * 0,098965$

$P_i = 0,098965$ Kg.m/s

$P_f = m_f V_{1f}$

$= 0,39 * 0,30314$

$P_f = 0,11822$ Kg.m/s

$\Delta P = P_f - P_i$

$= 0,11822 - 0,098965$

$\Delta P = 0,02$ Kg.m/s ≈ 0

2. Is the kinetic energy conserved? Explain that numerically.

No, it isn't.

$KE_i = \frac{1}{2} m_i v_i^2$

$= \frac{1}{2} * 0.195 * (0.507511)^2$

$KE_i = 0.025113\ J$

$KE_f = \frac{1}{2} m_f v_f^2$

$= \frac{1}{2} * 0.39 * (0.30314)^2$

$KE_f = 0.017919\ J$

$\Delta KE = KE_f - KE_i$

$= 0.017919 - 0.025113$

$\Delta KE = -0.007194 \neq 0$

Conclusion:

1. A collision occurs when two bodies come in direct contact and exert force on each other.

2. In the elastic collisions in this experiment, energy and momentum was overall only somewhat conserved.

Experiment #9: Simple Harmonic Motion

(Hooke's Law)

Data and Data Analysis:

Part (A): Determination the spring constant k by measuring elongation versus applied force.

عدا ابو بي

Table (1): x_0= ....0,7 m...., M= .....

m (g)	x (m)	$\Delta x = x - x_0$ (m)	F = mg (N)	K (N/m)
60	0,58	0,12	0,5886	4,905
80	0,48	0,22	0,7848	3,5673
100	0,45	0,25	0,981	3,924
120	0,42	0,28	1,1772	4,2043
			k= 4,15015	

1. Plot a graph of F versus (Δx).
2. Find the slope of the best straight line? What does the slope represent?
3. Calculate the average value of the spring constant $\bar{k}$ and standard deviation of the mean value Δk.
4. Write down the answers to parts (2-3) in the Results Section below

عدا ابو بي

Results:

$$F = K\,\Delta X$$

$$K = \frac{F}{\Delta X} \Rightarrow \text{slope} = \frac{F}{\Delta X} = K$$

slope = K

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{1,1772 - 0,5886}{0,28 - 0,12}$$

Slope = 3,67875.

$$\bar{K} = \frac{4,905 + 3,5673 + 3,924 + 4,2043}{4}$$

$\bar{k}$ = 4,15015 N/m.

$$= \sqrt{\frac{(4,15015 - 4,905)^2 + (4,15015 - 3,5673)^2}{5-1}}$$

$$= 0,476$$

$-1 < 0,476 < 1$ ✓

Part (B): Determination the spring constant k of a spring from measurements of the period T of oscillation.

Table (2)

m (Kg)	Time of (10) oscillation t (s)	τ (s)	τ^2 (s^2)
0,09	7.81	0,781	0,61
0,1	8,18	0,818	0,67
0,11	8.47	0,847	0,72
0,12	8,86	0,886	0,78

1. Plot τ^2 versus the load mass (m).
2. Find the slope of the best straight line.
3. Use the calculated slope to find k.
4. Compare your result with the value of k found in part A.
5. Write down the answers to parts (2-3) in the Results Section below

Results

$$\frac{\tau^2}{m} = \frac{4\pi^2}{K} \Rightarrow \text{slope} = \frac{4\pi^2}{K}$$

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{0,78 - 0,61}{0,12 - 0,09}$$

slope = 5

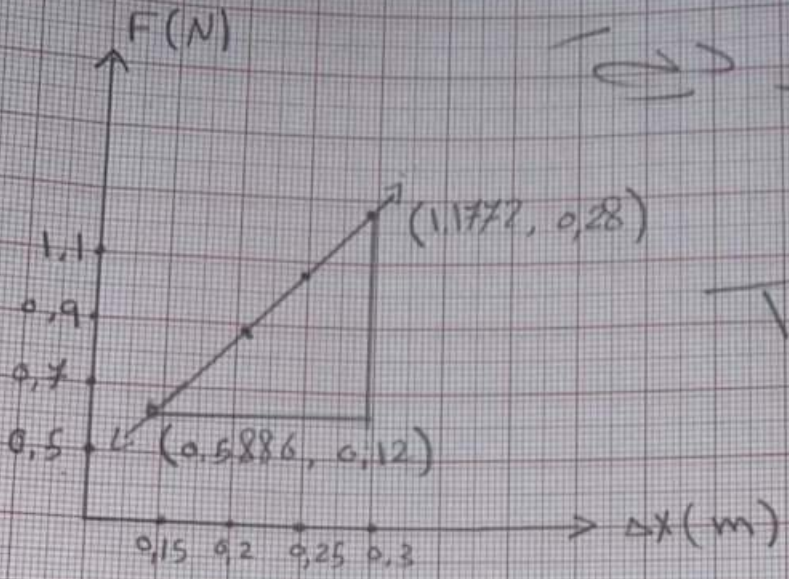
$$K = \frac{4\pi^2}{\text{slope}} = \frac{4 * (3.14)^2}{5}$$

$K = 6.96\ N/m$

Conclusion

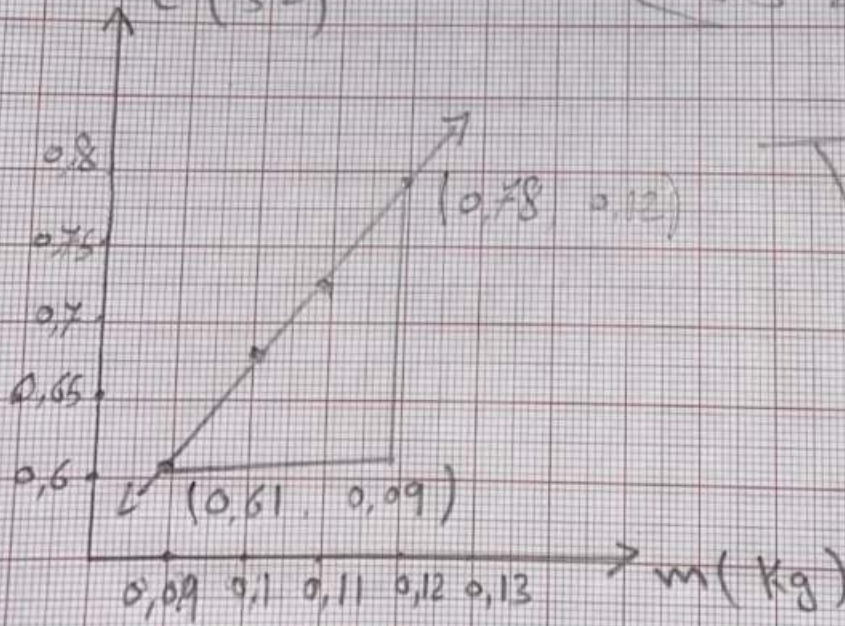
1. There is a linear relationship between the Force exerted on the spring and the amount of stretch or compression in the spring.

2. Force and spring extension are directly proportional related.



جدول (1)

Table (1)



جدول (2)

Table (2)

Experiment #8: Simple Harmonic Motion

(The Simple Pendulum)

Data and Data Analysis:

Part (A): Determining an experimental value of the acceleration due to gravity g.

Table (1): the mass of the bob = ... (Kg)

L (m)	Time of (10) oscillations t (s)	τ (s)	τ^2 (s^2)
0,2	8,97	0,897	0,8046
0,3	10,99	1,099	1,2078
0,4	12,69	1,269	1,6104
0,5	14,19	1,419	2,01356

1. Calculate τ the period of one oscillation.
2. Calculate τ^2.
3. On the graph paper, plot τ^2 versus L.
4. Calculate the slop and find g.
5. Calculate the percentage error for g.
6. From your graph determine the period of a simple pendulum 30 cm length.
7. From your graph determine the length of pendulum which has a period of 1 second.
8. Write down the answers to parts (4-7) in the Results Section below

Results:

(4) $slope = \frac{g}{4\pi^2} = \frac{L}{\tau^2}$

$slope = \frac{\Delta y}{\Delta x}$

$= \frac{0,5 - 0,2}{2,01356 - 0,8046}$

slope = 0,248

$g = slope * 4\pi^2$

$g = 9,78\ m/s^2$

(5) $P.E = \left|\frac{g_{ac} - g}{g}\right| * 100\%$

P.E = 2 %.

(6) T = 0,3 m

τ = 1,099 s

Part (B): Investigation the dependence of the period T of a pendulum on the length L and the mass M of the bob.

Table (2): L = ...0,3..... (m)

M (Kg)	Time of ( ) oscillations t (s)	τ (s)	$f=\frac{1}{\tau}$ (Hz)
Wood	9,93	0,993≈1	1,007
AL	8,73	0,873≈1	1,145
Fe	10,96	1,096≈1	0,9124

1. Calculate τ the period of one oscillation.
2. Calculate the frequency f. Tabulate your answers in Table 2 above
3. What is the effect of the mass on the periodic time? No Effect.

Conclusion

1. Only the length effects the period of a pendulum.

2. The mass of the pendulum has no relation to periodic time.

الرسم البياني

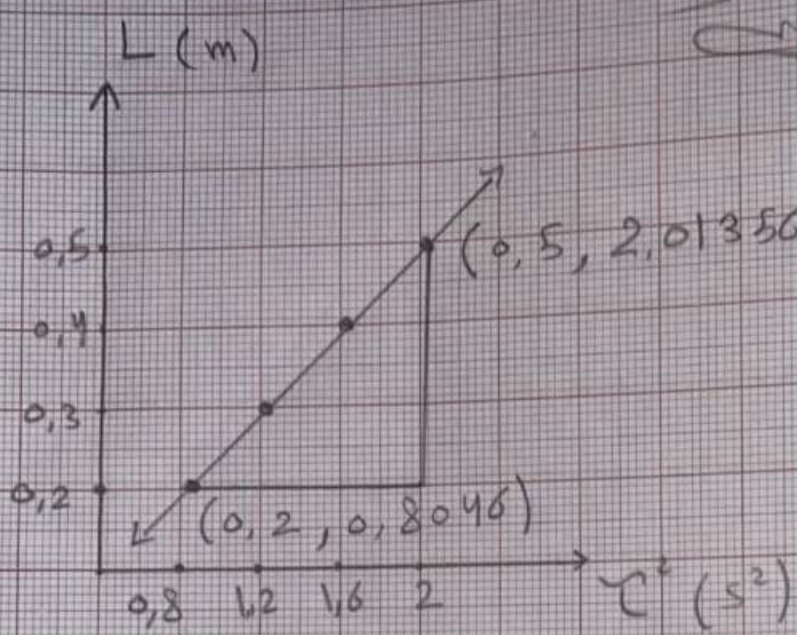


Table (1)

الرسم البياني

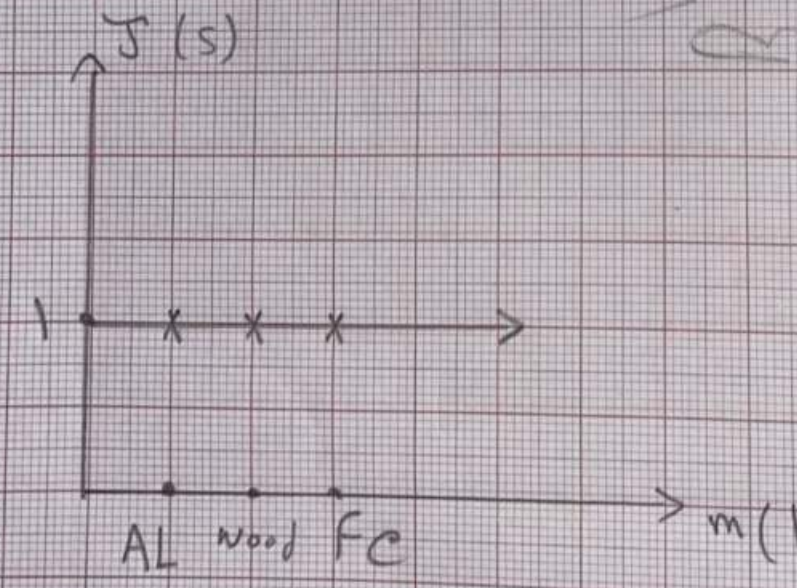


Table (2)